

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Spring-2025

Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline Course Code: MDP-3604

Total Marks-50

Time: 2:30 Hours

=====

Answer the following questions; Figures in the right hand margin indicate full marks

1. a) Deduce the Islamic attitudes towards sciences. 2
b) What is the source of knowledge in Islamic view? 2
c) Cite about the challenges of the Holy Quran 2
d) What are the dominant elements of the universe? 2
e) In which galaxy is our solar system located? 2
2. a) "Astronomers have noticed that all the stars are moving away from each other". What is the future of the universe (Open Universe or Closed Universe or Flat Universe)? Discuss in the perspective of Science and Quran. 6
b) What lessons can be learned from the Quran about creatures like bees and ants? 4
3. a) What do you mean by "Nutfah" and "Sulalah"? 2
b) Describe the stages of fetal development in the womb in the light of the Quran and modern embryology. 8
4. a) Observe your surrounding environment (home, neighborhood, or campus) and identify at least two ways in which environmental principles mentioned in Islam (such as the representation of nature or kindness to animals) are not being observed or violated. Present your findings in a short presentation or poster. 5
b) Analyze the Qur'an, Hadith, and Scientific Presentations on "Children Will Be Sons or Daughters" 5
OR
4. What are the negative impacts of having sex beyond of marriage, free mixing, extramarital affair, homosexuality, pornography and short dress on society? 10
5. a) Classify disasters with examples 2
b) Mention three examples of natural disasters described in the Qur'an 3
c) The Messenger of Allah said Natural Disaster occurs due to fourteen crimes. What are those? 5
OR
5. Keep a 3-day food and activity diary. Consider how your lifestyle is in line with Islamic guidelines regarding health, diet, and cleanliness. Make suggestions for improvement based on Islamic teachings. 10

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring-2025

Course Code: URED-3604 Course Title: Life and Teachings of Prophet (SAAS)
(For Law faculty: URED-3201)

Full Marks: 50

Time: 2:30 Hours

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Discuss the background of the <i>Hijrah</i> of the Prophet Muhammad (SAAS) from Makkah to Madinah and assess its significance as a turning point in the history of Islam.	10	3	Understand & Evaluate
2	a) Analyze the causes of the Battle of Badr, mentioning the key events and the lessons drawn from it. Or, b) Examine the background of the Battle of Ahzab, and analyze the strategic measures taken to protect Madinah. Discuss the major events of the battle and the lessons it offers for the Muslim community.	10	3	Remember, Analyze & Create
3	"The Treaty of Hudaibiyyah is described as <i>Fathum Mubeen</i> (clear victory) in the holy Qur'an"- evaluate why is it considered so?	10	3	Evaluate
4	"Makkah was conquered without any bloodshed"- explain how it was achieved from a historical point of view.	10	3	Create
5.	Critically evaluate the Farewell Sermon of Prophet (SAAS) as a foundational charter of human rights in the history of mankind.	10	3	Evaluate

=====

International Islamic University Chittagong
Center for General Education (CGED)

Semester End Examination: Spring 2025

Course Code: GEHE-3601

Program: Undergraduate

Course Title: History of Emergence of Bangladesh

Full Marks: 50

Time: 2 hours and 30 minutes.

Instructions:

- i. All Questions are Compulsory.
- ii. Figures in the right margin indicate full marks.
- iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.

Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	What is nationalism? Discuss the origin and development of Bengali nationalism. How did Bengali nationalism evolve into Bangladeshi nationalism? Or Assess the historical context of the formation of the United Front and investigate the key factors of its success in the election of 1954.	10	R	CLO2
2	Analyze the Six-Point Demands of 1966 as a 'firm solution of Pakistan's inter-wing political and economic problem'. How did these demands contribute to the political awakening and eventual emergence of Bangladesh?	10	An	CLO3
3	What were the major political, economic, and cultural factors that led to the overwhelming victory in the elections of 1970? How did it become a turning point in the history of Bangladesh?	10	U	CLO3
4	Evaluate the responses and roles of superpowers in the Liberation War of Bangladesh within the context of the contemporary Cold War.	10	E	CLO3
5	Write short notes on any two: a) Agartala Conspiracy Case of 1968 b) Eleven Sectors during the Liberation War of 1971 c) 4 th Amendment of the Constitution.	5 5	E	CLO3

Course Code: ECON 3501

Course Title: Principle of Economics

Time: 2 hours 30 minutes

Full Marks: 50

- (i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions			
CO	CO Statements	PO	BL
CO1	Explain the knowledge of the fundamental concepts and theories of micro and macro-economics.	PO11	C1,C2
CO2	Analyze the key indicators of economic growth.	PO11	C4
CO3	Compare the economic theories and concepts to analyze behavior of individuals, firms and nations to act as a responsible citizen.	PO11	C5

Bloom's Levels (BL) of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group- A

[Answer the questions from the followings]

1. a) Company X and Company Y are the only two producers of smartphones in a country. Company X lowers its price by 10%. The next day, Company Y does the same. CO2 C2 5
What kind of market structure is this and how does it differ from perfect competition market?
1. b) Explain what "interdependence" means in this context. Why might prices in this market be stable even with competition? CO2 C2 5
2. a) Several market structures are perform in the world, what are they and what can it tell us about how a firm will behave? CO1 C4 5
2. b) The table below shows the market demand schedule and the cost structure. CO2 C5 5

P	Q	Fixed Cost	Variable Cost	TC	AVC	MC
20	1	80 TK.	200 TK.			
18	2	80 TK.	280 TK.			
16	3	80 TK.	375 TK.			
14	4	80 TK.	400 TK.			
12	5	80 TK.	500 TK.			

Measure TC , ATC, AVC and MC for each row.

Or

2. a) What do you mean by 'GDP Growth'? Distinguish between Nominal and Real GDP with example. CO2 C4 5
2. b) GDP includes final goods and services only. Why? Describe the Value-Added Approach of measuring GDP with example. CO3 C4 5

Group- B

[Answer the questions from the followings]

3. a) Point out the relation between population growth and economic growth. CO2 C5 5
3. b) Economic plans often include goals such as full employment, self-sufficiency, and industrialization. How do these goals reflect the broader challenges of economic development in underdeveloped countries? CO3 C5 5

OR

- | | | | | | |
|----|----|---|-----|----|---|
| 3. | a) | Critically distinguish between economic growth and economic development. Why is it important for policymakers in developing countries to focus on development rather than just growth | CO2 | C5 | 5 |
| 3. | b) | Discuss Harrod-Doman growth models for developing countries. | CO1 | C2 | 5 |
| 4. | a) | GDP per person tells us the income and expenditure of the average person in an economy. But it does not reflect the happiness or quality of life. Why, explain. | CO2 | C4 | 5 |
| 4. | b) | The government of a country is not the only responsible entity for economic development. The citizens of the country also have some responsibilities. Write some of such responsibilities that you can do as a citizen. | CO2 | C1 | 5 |
| 5. | a) | Discuss the roles of monetary and fiscal policy in stabilizing the economy during a recession. | CO1 | C2 | 5 |
| 5. | b) | Briefly explains the objectives of Trade Policy in Bangladesh. How Trade Policy works positively? | CO1 | C2 | 5 |

Course Code: CSE-3635

Course Title: Artificial Intelligence

Total Marks: 50

Time: 2 Hours 30 Minutes

[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a separate answer script for Group-A and Group-B.]

Group A

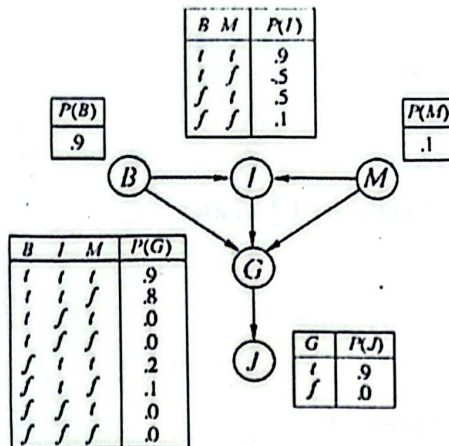
	Marks	COs
1. a) Define Inference. Determine whether the following sentence is i) Satisfiable ii) Contradictory iii) Valid $(P \vee Q) \rightarrow (P \wedge Q)$	3	2
b) State modus ponens with an example? Consider following sentences: Premise1: Every Bangladeshi likes Jackfruit. Premise2: Shihab is a Bangladeshi. Conclusion: Shihab likes Jackfruit. Why Propositional logic fails here?	3	2
c) Given knowledge base R1, R2, R3, R4, R5 below, show that P1,2 is false. R1: $\neg P1,1$ R2: $B1,1 \leftrightarrow (P1,2 \vee P2,1)$ R3: $B2,1 \leftrightarrow (P1,1 \vee P2,2 \vee P3,1)$ R4: $\neg B1,1$ R5: B2,1	3	2
d) What is the logical translation of the following statement? "None of my friends are perfect." (A) $\exists x(F(x) \wedge \neg P(x))$ (B) $\exists x(\neg F(x) \wedge P(x))$ (C) $\exists x(\neg F(x) \wedge \neg P(x))$ (D) $\neg \exists x(F(x) \wedge P(x))$	1	2
OR		
1. a) A statute mandates that it is an offense for a citizen of Nation X to furnish controlled substances to a nation designated as a prohibited recipient. The nation of Zargon is formally designated as a prohibited recipient by the government of Nation X. Zargon has recently operationalized a novel, high-capacity power generation system. All essential rare-earth elements required for the construction of this power generation system were supplied by Ms. Anya, who holds citizenship in Nation X. Prove by that Ms. Anya Sharma has committed an offense by both - 1) Forward chaining 2) Backward chaining	6	2
b) Describe the following concepts with example- 1) Generalized Modus Ponens 2) Hypothetical Syllogism	4	2
2 a) Why does first-order predicate logic fail to adapt to a domain like medical diagnosis? How can agents manage the uncertainty raised by a lack of environmental knowledge?	3	3

- b) A domain consisting of just the three Boolean Variables Toothache, Cavity and Catch. The full joint distribution is a $2 \times 2 \times 2$ table as shown in figure.

	Toothache		$\neg$ Toothache	
	Catch	$\neg$ Catch	Catch	$\neg$ Catch
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

Calculate:

- $P(\text{Cavity} \vee \neg \text{Catch})$
 - $P(\neg \text{Cavity} | \text{Toothache}, \neg \text{Catch})$
 - $P(\text{Cavity} | \neg \text{Toothache})$
 - $P(\text{Toothache} | \text{Cavity})$
- c) A Bayesian network with Boolean variables $B = \text{Broke Election Law}$, $I = \text{Indicted}$, $M = \text{Politically Motivated Prosecutor}$, $G = \text{Found Guilty}$, $J = \text{Jailed}$, showing both the topology and the conditional probability tables.



- Calculate the value of $P(b, i, m, g, j)$
- Calculate the probability that someone goes to jail given that they broke the law, have been indicted, and face politically motivated prosecutor.

Group B

- Define conditional probability. Write the difference between degree of belief and degree of truth with example.
 - "Fuzzy set is the generalized version of crisp set" - Do you agree with the statement? Justify your answer
 - Given the following membership functions for the fuzzy set "Temperature":

$$\mu_{\text{Cold}}(x) = \begin{cases} 1, & x \leq 10 \\ \frac{20-x}{10}, & 10 < x < 20 \\ 0, & x \geq 20 \end{cases}$$

Find the degree of membership for a temperature of 15°C in the set "Cold".

- A doctor knows that meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is $1/50,000$, and the prior probability that any patient has a stiff neck is $1/20$. Find the probability of patients with stiff neck having Meningitis.

OR

- Describe the concept of inductive learning with an example. Is reinforcement learning considered a subset of inductive learning? Justify your answer.

- b) Write down both the top down and bottom up parse tree for the sentence: "Nusair created the program" using the following simple grammar for a fragment of English in Figure 1. 6 3

$S \rightarrow NP VP$
 $NP \rightarrow the NPI$
 $NP \rightarrow PRO$
 $NP \rightarrow PN$
 $NP \rightarrow NPI$
 $NPI \rightarrow ADJS N$
 $ADJS \rightarrow \epsilon \mid ADJS ADJ$
 $VP \rightarrow V$
 $VP \rightarrow V NP$
 $N \rightarrow file \mid program$
 $PN \rightarrow Nusair$
 $PRO \rightarrow \epsilon$
 $ADJ \rightarrow short \mid long \mid fast$
 $V \rightarrow printed \mid created \mid wanted$

Figure 1: A simple grammar for a fragment of English

4. a) Write down the steps through which NLP is conducted? Why is sequencing important in NLP? 3 4
- b) What are the key components of an expert system? Explain with a diagram. 3 3
- c) Imagine you want to create an 'Intelligent Recipe Recommender' system. This system should act like an expert chef, suggesting recipes and giving cooking advice. How would you use Natural Language Processing (NLP) and Expert System ideas together to build this? Explain each step. 4 3
5. a) What is supervised learning? Explain the concepts of forward propagation and back propagation in neural networks. 4 4
- b) Consider a single neuron with two inputs X_1 and X_2 , and output Y , learning the logical AND operation. The truth table is given below. 6 4

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Develop an ANN model using Delta rule, Where,
 Threshold, $T = 0.2$; Learning rate, $\alpha = 0.2$; Initial weights $W_1=0.3$ and $W_2=-0.1$

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Final Examination, Spring 2025

Course Code: CSE-3525

Course Title: Data Communication

Total Marks: 50

Time: 2 Hours 30 Minutes

[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

- | | M | C | D |
|--|---|---|---|
| a) What do you mean by data rate and signal rate? Suppose you need to send data 2 bits at a time at a bit rate of 4 Mbps using FSK. The carrier frequency is 100 MHz. Calculate the number of levels, the baud rate, and the bandwidth. shows the allocation of frequencies and bandwidth. | 5 | 3 | A |
| b) Four sources, each with a bit rate of 800 kbps, need to be combined using Time Division Multiplexing (TDM) with a byte interval and synchronizing bits. Answer the following questions regarding the multiplexing process:
i. What is the size of a frame in bits?
ii. What is the frame rate?
iii. What is the duration of a frame?
iv. What is the data rate? | 5 | 3 | A |
| 2. a) Using the data "00100111", draw the waveform for QPSK techniques, where you should draw two composite signals and then add them to create the QPSK signal. Draw also the implementation diagram of this modulation technique. | 5 | 2 | A |
| b) Draw and explain the constellation diagram for ASK (OOK), BPSK, QPSK, and 8QAM signals. | 5 | 2 | U |
| OR | | | |
| a) Explain Frequency hopping spread spectrum with figure. | 5 | 2 | A |
| b) Draw and explain the analog hierarchical systems used by telephone companies is made up of groups, super groups, master groups, and jumbo groups. You should mention the required channel and bandwidth for each category. | 5 | 2 | U |

Group B

- | | | | |
|--|----|---|---|
| a) Explain Go back N protocol. Why is it inefficient? | 5 | 2 | A |
| b) Explain Bit stuffing and Byte stuffing. Show Bit stuffing and unstuffing using 000111110011111101000 data. | 5 | 2 | A |
| a) Draw the block diagram of the checksum error detection technique.
Block 1: 10011001 Block 2: 11100010
Block 3: 00100100 Block 4: 10000100
Consider these data blocks and perform the checksum error detection technique for both the sender and receiver sides. | 5 | 3 | A |
| b) Given the dataword 1010011110 and the divisor 10111, show:
The codeword generation at the sender site
The codeword checking at the receiver site, assuming no error occurred.
We have 3 users in a CDMA system. Each user sends their own data, which is spread using a unique Walsh code before transmission. The information for each user is given below:
 User Data Walsh Code
 ----- ----- -----
 User 1 00 0101
 User 2 10 0011
 User 3 11 0000 | 5 | 3 | A |
| a) Perform spreading for each user and write the resulting 8-chip spread signals.
b) Plot all users' spreading signals and compute the combined CDMA signal by summing the spread signals of all users.
c) Plot the combined signal waveform.
d) Describe and plot how a receiver can extract User 2 and 3's original data from the combined signal. | 10 | 4 | U |
| OR | | | |
| a) Suppose you have a 4-bit dataword 1100 and you are using Hamming Code C(7,4) for error correction. Show the codeword at the sender site. If the data word is changed to 1000, show the codeword received. How does the receiver identify the position of the erroneous bit? | 5 | 4 | U |
| b) Explain Selective-Repeat ARQ with proper diagrams. | 5 | 4 | U |

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE, Semester Final Examination

Session: Spring-2025
 Course Code: CSE-3631
 Time: 2 hour 30 Minutes

Semester: 6th
 Course Title: Operating System
 Marks: 50

- (i) The figures in the right-hand margin indicate full marks
 (ii) Course Outcomes is mentioned in additional Columns

Course Learning Outcomes (CLOs):	
CLO1	Explain the fundamental principles and core services provided by an operating system
CLO2	Analyze and evaluate scheduling algorithms for process management and synchronization mechanisms to ensure efficient resource allocation.
CLO3	Evaluate and apply memory management techniques, virtual memory concepts, file system structures and disk management techniques to optimize resource utilization.

Group A

**CLO M
ar
ks
10**

Q1

A system has 4 processes and 5 allocate resource. The current allocation and maximum needs are as follows-

	Allocation					Maximum				
	A	B	C	D	E	A	B	C	D	E
P1	1	0	2	1	1	1	1	2	1	3
P2	2	0	1	1	0	2	2	2	1	0
P3	1	1	0	1	1	2	1	3	1	1
P4	1	1	1	1	0	1	1	2	2	0

Using Banker's algorithm, answer the following questions: -

- What are the contents of need matrix?
- If initial value = $[0 \ 0 \ X \ 1 \ 1]$, what is the smallest value of x for which this is a safe state? find the safe sequence.
- If a request from process P3 arrives for (0 0 1 1 1), can the request be granted immediately? If yes find the safe stat.

Or

Q1 a)

A virtual memory system employs a main memory with 4-page frames. Consider the following page reference string generated by the CPU: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3. Demonstrate the application of the Optimal Page Replacement Algorithm for the given page reference string and frame size. For each page reference, clearly indicate whether a page fault (F) or a page hit (H) occurs. Calculate and state the total number of page faults incurred. Justify your eviction choices at each step where a page replacement is necessary by referring to the future page references.

b)

A foundational concept in operating systems is efficient memory management, particularly when physical memory is limited. Assume a system is configured with a main memory comprising 3-page frames. The CPU generates the following page reference string: 1, 3, 0, 3, 5, 6, 3. Applying the First-In, First-Out (FIFO) Page Replacement Algorithm, systematically trace the page replacements for the given reference string and frame configuration. For each

page reference processed, explicitly denote whether it results in a page fault (F) or a page hit (H). Calculate and present the total number of page faults observed throughout the entire sequence.

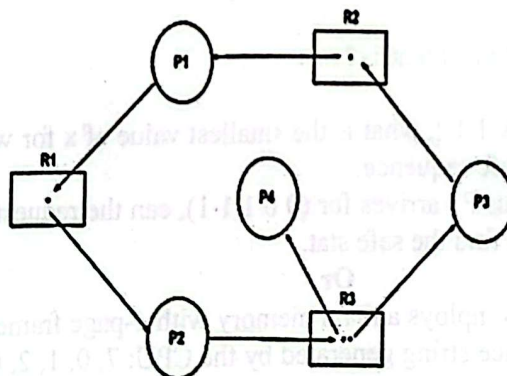
- Q2**
- | | | | |
|----|---|---|---|
| a) | Deadlock represents a critical state within a multi-programmed operating system where a set of processes becomes permanently blocked, each holding a resource and simultaneously waiting for a resource held by another process in the set. Analyze and articulate the necessary conditions that must simultaneously hold true for a deadlock to arise in a system. For each condition, explain its individual contribution to the deadlock scenario. | 2 | 3 |
| b) | What are the key factors that influence the performance of demand paging? Explain the concept of the Effective Memory Access Time (EMAT) in a paged system. How is it calculated? | 2 | 4 |
| c) | Discuss the potential drawbacks and challenges associated with deadlock recovery | 3 | 3 |

Group B

- Q3**
- | | | | |
|----|--|---|---|
| a) | Consider a swapping system in which memory consists of the following hole sizes in memory order: 15KB, 8KB, 113KB, 10KB, 7KB and 18KB. Which hole is taken for successive segment requests (Representation by diagram is preferable):
i. 113 KB ii. 115 KB iii. 7 KB iv. 20 KB v. 7 KB for the first fit, best fit and worst fit? Which makes the most efficient use of memory? | 3 | 4 |
| b) | In spite of limited physical memory we can execute of program whose logical memory size is greater than physical memory of the system. How it's possible? | 2 | 3 |
| c) | Describe steps in handling a page fault with proper diagram. | 2 | 3 |

Or

Q3



- | | | | |
|----|---|---|---|
| a) | Consider the above-mentioned multiple instances resource allocation graph for determining if the state is safe or not and also provide an allocation and request matrix to justify the outcome. | 2 | 5 |
| b) | Deadlock represents a critical state within a multi-programmed operating system where a set of processes becomes permanently blocked, each holding a resource and simultaneously waiting for a resource held by another process in the set. Analyze and articulate the necessary conditions that must simultaneously hold true for a deadlock to arise in a system. For each condition, explain its individual contribution to the deadlock scenario. | 2 | 5 |

- Q4**
- | | | | |
|----|--|---|---|
| a) | A virtual memory system employs a main memory with 4 page frames. Consider the following page reference string generated by the CPU: a, b, c, d, e, e, b, f, c, b, c, d, b, a, b, c. Demonstrate the application of the Optimal Page Replacement | 2 | 5 |
|----|--|---|---|

Algorithm for the given page reference string and frame size. For each page reference, clearly indicate whether a page fault (F) or a page hit (H) occurs. Calculate and state the total number of page faults incurred. Justify your eviction choices at each step where a page replacement is necessary by referring to the future page references.

- b) Virtual memory, a cornerstone of modern operating systems, fundamentally abstracts the concept of physical memory, allowing programs to operate as if they have access to a very large address space. This abstraction is commonly realized through demand paging, a technique where processes reside in secondary memory, and pages are loaded into main memory only when they are actively referenced, rather than pre-loaded in their entirety. Elaborate on the core mechanism of a demand paging system. Explain how it leverages the principles of virtual memory and paging with swapping to enable execution of programs larger than physical memory. 2 5

- Q5 a) Why does thrashing occurs? Explain the working set model in thrashing. 2 3
- b) What is file allocation method? Describe contiguous and Linked allocation method. 2 2
- c) Consider an access matrix for a system with three subjects (S1, S2, S3) and four objects (O1, O2, O3, O4). Each entry in the matrix represents the access rights for a subject to an object, where 0 means no access, 1 means read access, and 2 means write access. The access matrix is as follows: 3 5

	O1	O2	O3	O4
S1	1	2	0	1
S2	0	1	1	0
S3	2	0	0	1

Calculate the total number of access rights granted in the access matrix.

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B. Sc (Engg.) in CSE Semester: 6th

Final-Term Examination, Spring 2025

Course Code: CSE-3637

Course Title: Software Engineering

Time: 2 hours and 30 minutes

Total Marks: 50

[Answer all the questions. Use separate answer scripts for Group A and Group B Figures in the right-hand margin indicate full marks.]

Group A

Marks CO

1. Read the passage and answer questions 1(a-c):

Let's consider, you are managing the development of a new hospital management system (HMS) that is designed to streamline operations, improve patient care, and ensure compliance with healthcare regulations. The system will include modules for patient registration, appointment scheduling, billing, medical records and reporting.

As the project manager, you need to identify and categorize the stakeholders involved in the project to ensure that their needs and concerns are addressed throughout the development process.

- | | | | |
|-----|--|---|-----|
| (a) | Identify the primary, secondary, and tertiary stakeholders for the hospital management system (HMS). | 3 | CO2 |
| (b) | Explain the roles and interests of each category of stakeholders in the project. | 4 | CO2 |
| (c) | Describe how you would engage each group of stakeholders to ensure the project's success. | 3 | CO2 |

OR

1. Read the passage and answer the questions Q1(a-c):

You are assigned to develop an Online Medical Appointment System for a chain of private hospitals. The system must allow patients to register, view doctor availability, book appointments, receive notifications, and submit feedback. Doctors will manage their consultation schedules and access patient records. Hospital administrators will review booking statistics, and IT staff will maintain system performance.

- | | | | |
|----|---|---|-----|
| a) | Identify at least four stakeholders involved in this system. Categorize them according to stakeholder types (Primary, Secondary, Tertiary) and justify your classification. | 3 | CO2 |
| b) | For the given system, draw a viewpoint hierarchy diagram. Explain how each viewpoint helps in identifying distinct requirement clusters. | 4 | CO2 |
| c) | Based on this scenario, identify two key functionalities and illustrate them using a use case diagram. | 3 | CO2 |

2 Read the passage and answer the questions Q2(a-c):

You have been assigned to design a School Management Software that will manage student admissions, course registrations, teacher assignments, exam schedules, and online result publication. The system should be modular so that changes in one function (e.g., course registration) do not affect others. It must also be expandable to support features like online class tracking and attendance monitoring in future phases.

- | | | | |
|-----|--|---|-----|
| (a) | Based on the given scenario, identify any four key software design concepts. | 4 | CO3 |
| (b) | Draw a high-level architecture diagram representing how different modules (admission, course, exams, etc.) are structured and communicate with each other.
Explain how the principles of low coupling and high cohesion can be applied to improve the maintainability of this system | 6 | CO2 |

Group B

- | | | | |
|------|---|---|-----|
| 3(a) | A beta version of a mobile payment app is released to internal staff. Describe how Alpha and Beta testing differ in process, participants, and outcome. Provide examples for both in this scenario. | 3 | CO3 |
| (b) | You are part of a QA team for a hospital appointment system. The client reports that the confirmation button does not respond sometimes. Distinguish between verification and validation in this context. Suggest one technique for each that could help resolve this. | 2 | CO4 |
| (c) | A logistics app crashes under high load. Recommend at least three types of non-functional testing that would help identify the issue and explain how each would improve reliability and user satisfaction. | 5 | CO3 |
| 4(a) | From a software maintenance perspective, why is documentation vital? Discuss this with examples from any projects. | 3 | CO3 |
| 4(b) | Explain the main causes of software maintenance challenges and suggest potential solutions. | 3 | CO2 |
| (c) | A manager asks you to explain the software reengineering process to stakeholders from marketing and finance.
Prepare a short, jargon-free presentation of this clearly.
OR | 4 | CO3 |
| (a) | Describe the necessary steps in the COCOMO model. | 5 | CO2 |
| (b) | The fan in and fan out of module X is 3 and 4 respectively. The complexity of the system is 2000. The number of lines in module X is 200, i.e LOC of module X is $LOC(X) = 200$. Calculate the structural complexity of module X using card and glass's system complexity. Using combined Henry Kafura's approach and Card glass's approach calculate the data complexity of module X. | 5 | CO3 |

5. Read the passage and answer questions 5(a-c):

You are the project manager of a software development company that has been operating at CMM Level 2 (Repeatable) for the past two years. Your company has been successful in maintaining its project schedules and budget, and there is a set process for project management and quality assurance. However, the company has faced challenges with process variability, inconsistent project performance, and limited process improvement initiatives.

Your CEO wants to move the company to CMM Level 3 (Defined) within the next year to achieve more consistent project outcomes and to enhance the company's competitive edge. To do this, you need to implement standardized and documented processes across the organization.

- | | | | |
|-----|--|---|-----|
| (a) | What key steps should you take to transition your company from CMM Level 2 to CMM Level 3? | 3 | CO2 |
| (b) | What challenges might you encounter during this transition, and how would you address them? | 4 | CO2 |
| (c) | How will your company reach CMM Level 3 benefit in terms of project performance and customer satisfaction? | 3 | CO2 |